

T0-Model: Complete Parameter-Free Particle Mass Calculation Direct Geometric Method vs. Extended Yukawa Method With Complete Neutrino Quantum Number Analysis and QFT Derivation

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Abstract

The T0-Model offers two mathematically equivalent but conceptually different calculation methods for particle masses: the direct geometric method and the extended Yukawa method. Both approaches work with the single geometric constant $\xi = \frac{4}{3} \times 10^{-4}$; the rational quantum numbers (r_i, p_i) are structural choices, and ν serves as a conversion anchor. This documentation also contains neutrino quantum numbers and the quantum field theoretical derivation of the ξ -constant through EFT matching and 1-loop calculations. The neutrino part with double ξ -suppression is not tenable according to R109 (see the addendum at the end); for neutrinos, Doc. 340 is authoritative. For charged leptons and quarks, the average deviation is below 1%; the over twenty free Standard Model mass parameters are thereby traced back to the single parameter ξ and fixed rational quantum numbers.

.1 Introduction

Particle physics faces a fundamental problem: The Standard Model with its over twenty free parameters offers no explanation for the observed particle masses. They are measured as inputs, not derived from a deeper principle. The T0-Model addresses this with two complementary calculation methods that work with the single parameter ξ and rational quantum numbers; the originally included treatment of neutrino masses is not tenable according to R109 (see addendum).

The Parameter Problem of the Standard Model

Despite its experimental success, the Standard Model has a theoretical gap: It contains more than 20 free parameters that must be determined experimentally. These include:

- **Fermion Masses:** 9 charged lepton and quark masses
- **Neutrino Masses:** 3 neutrino mass eigenvalues
- **Mixing Parameters:** 4 CKM and 4 PMNS matrix elements
- **Gauge Couplings:** 3 fundamental coupling constants
- **Higgs Parameters:** Vacuum expectation value and self-coupling
- **QCD Parameters:** Strong CP phase and others

Parameter Reduction

The T0-Model traces the over twenty free parameters of the Standard Model back to **one**: the geometric constant $\xi = \frac{4}{3} \times 10^{-4}$, motivated by the geometry of three-dimensional space. In addition there are the rational quantum numbers (r_i, p_i) as

structural choices and ν as a conversion anchor. This version also contains neutrino quantum numbers (not tenable according to R109, see addendum) as well as the quantum field theoretical derivation.

.2 Methodological Clarification: Establishment vs. Prediction

Scientific-Historical Classification

The T0-Model draws on the methodology of **pattern recognition and systematic classification**, which was successful, for example, for the periodic table (Mendeleev 1869) or the quark model (Gell-Mann 1964).

Two-Phase Development

Phase 1: Establishment of Systematics

- 1. Pattern recognition in known particle masses (electron, muon, tau)
- 2. Parameter determination from experimental data
- 3. Establish quantum number assignment
- 4. Demonstrate mathematical equivalence of both methods

Phase 2: Unfolding Predictive Power

- 1. Extrapolation to unknown particles
- 2. Derive quark sector from lepton patterns
- 3. Predict new generations
- 4. Conduct experimental tests

Historical Precedence of Successful Pattern Physics

The comparison concerns the method of pattern recognition, not the state of confirmation:

Discovery		Pattern Recognition	Predictions	Confirmation
Periodic Table (1869)		Atomic weights and properties	Gallium, Germanium, Scandium	Experimentally confirmed
Spectral Lines (1885)		Hydrogen lines	Rydberg formula for all series	Quantum mechanics
Quark Model (1964)		Hadron masses	Eightfold Way	QCD theory
T0-Model (2025)		Lepton masses	Quarks	Experimental tests

Table 1: Historical precedence of pattern physics

.3 From Energy Fields to Particle Masses

The Fundamental Challenge

A central result of the T0-Model is that particle masses can be calculated from geometric principles. While the Standard Model requires over 20 free parameters to describe particle masses, the T0-Model works with the geometric constant $\xi_{\text{geom}} = \frac{4}{3} \times 10^{-4}$ and rational quantum numbers; the deviations for charged leptons and quarks are below 1.2%.

Parameter Reduction for Masses

Parameter Reduction:

- **Standard Model:** 20+ free mass parameters (measured)
- **T0-Model:** 1 parameter ξ plus rational quantum numbers (geometric)
- **Experimental Accuracy:** 99% average agreement (charged leptons and quarks; neutrino part see addendum)
- **Theoretical Foundation:** Three-dimensional spatial geometry + QFT derivation

Energy-Based Mass Concept

In the T0 framework, what we traditionally call "mass" is interpreted as an expression of characteristic energy scales of field excitations:

$$m_i \rightarrow E_{\text{char},i} \quad (\text{characteristic energy of particle type } i) \quad (1)$$

This reinterpretation treats mass and energy as different aspects of the same quantity.

.4 Two Complementary Calculation Methods

The T0-Model offers two mathematically equivalent but conceptually different approaches to calculating particle masses:

Method 1: Direct Geometric Resonance

Conceptual Foundation: Particles as resonances in the universal energy field

The direct method treats particles as characteristic resonance modes of the energy field $E(x,t)$, analogous to standing wave patterns:

$$\text{Particle} = \text{Discrete resonance modes of } E(x,t)(x,t) \quad (2)$$

Three-Step Calculation Process:

Step 1: Geometric Quantization

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i) \quad (3)$$

Here $f(n, l, j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$.

where:

$$\xi_0 = \frac{4}{3} \times 10^{-4} \quad (\text{geometric base parameter}) \quad (4)$$

$$n_i, l_i, j_i = \text{quantum numbers from 3D wave equation} \quad (5)$$

$$f(n_i, l_i, j_i) = \text{geometric function from spatial harmonics} \quad (6)$$

Step 2: Resonance Frequencies

$$\omega_i = \frac{c^2}{\xi_i \cdot r_{\text{char}}} \quad (7)$$

In natural units ($c = 1$):

$$\omega_i = \frac{1}{\xi_i} \quad (8)$$

Step 3: Mass Determination from Energy Conservation

$$E_{\text{char},i} = \hbar \omega_i = \frac{\hbar}{\xi_i} \quad (9)$$

In natural units ($\hbar = 1$):

$$E_{\text{char},i} = \frac{1}{\xi_i} \quad (10)$$

Method 2: Extended Yukawa Method

Conceptual Foundation: Bridge to Standard Model formulation

The extended Yukawa method maintains compatibility with Standard Model calculations while making Yukawa couplings geometrically determined rather than empirically fitted:

$$E_{\text{char},i} = y_i \cdot v \quad (11)$$

where $v = 246$ GeV is the Higgs vacuum expectation value (SM definition: $v \equiv (\sqrt{2} G_F)^{-1/2}$; no independent observable, Doc. 351/R112).

Geometric Yukawa Couplings:

$$y_i = r_i \cdot \left(\frac{4}{3} \times 10^{-4} \right)^{\pi_i} \quad (12)$$

Generation Hierarchy:

$$\text{1st Generation: } \pi_i = \frac{3}{2} \quad (\text{electron, up quark}) \quad (13)$$

$$\text{2nd Generation: } \pi_i = 1 \quad (\text{muon, charm quark}) \quad (14)$$

$$\text{3rd Generation: } \pi_i = \frac{2}{3} \quad (\text{tau, top quark}) \quad (15)$$

The coefficients r_i are simple rational numbers determined by the geometric structure of each particle type.

.5 Quantum Field Theoretical Derivation of the ξ -Constant

EFT Matching and Yukawa Coupling after EWSB

After electroweak symmetry breaking, we have the Yukawa interaction:

$$\mathcal{L}_{\text{Yukawa}} \supset -\lambda_h \bar{\psi} \psi H, \quad \text{with} \quad H = \frac{v+h}{\sqrt{2}} \quad (16)$$

After EWSB:

$$\mathcal{L} \supset -m \bar{\psi} \psi - y h \bar{\psi} \psi \quad (17)$$

with the relations:

$$m = \frac{\lambda_h v}{\sqrt{2}} \quad \text{and} \quad y = \frac{\lambda_h}{\sqrt{2}} \quad (18)$$

The local mass dependence on the physical Higgs field $h(x)$ leads to:

$$m(h) = m \left(1 + \frac{h}{v} \right) \Rightarrow \partial_\mu m = \frac{m}{v} \partial_\mu h \quad (19)$$

T0 Operators in Effective Field Theory

In T0 theory, operators of the form appear:

$$O_T = \bar{\psi} \gamma^\mu \Gamma_\mu^{(T)} \psi \quad (20)$$

with the characteristic time field coupling term:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} \quad (21)$$

Substituting the Higgs dependence:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} = \frac{1}{mv} \partial_\mu h \quad (22)$$

This suggests that a $\partial_\mu h$ -coupled vector current is the UV origin.

1-Loop Matching Calculation

The 1-loop amplitude for the T0 vertex yields:

$$F_V(0) = \frac{y^2}{16\pi^2} \left[\frac{1}{2} - \frac{1}{2} \ln \left(\frac{m_h^2}{\mu^2} \right) + \frac{r(r - \ln r - 1)}{(r-1)^2} \right] \quad (23)$$

For hierarchical masses ($m \ll m_h$), the constant term dominates:

$$F_V(0) \approx \frac{y^2}{32\pi^2} \quad (24)$$

ξ Formula from Higgs Physics

The EFT matching yields the relationship:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \quad (25)$$

With Standard Higgs parameters ($m_h = 125.1$ GeV, $v = 246.22$ GeV, $\lambda_h = m_h^2/(2v^2) \approx 0.129$):

$$\xi \approx 1.30 \times 10^{-4} \quad (26)$$

This lies close to the geometric setting $\xi_0 = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4}$ (deviation $\approx -2.4\%$). The relation is a consistency check of ξ , not an exact derivation. For the origin of the Higgs route see Doc. 385.

.6 Universal Particle Mass Systematics

Revised Universal Table of Fermions

Fermion	Generation	Family	Spin	r_f	Exponent p_f	SM Correspondence
Electron Neutrino	1	0	1/2	4/3	5/2	Double ξ
Electron	1	0	1/2	4/3	3/2	Lepton number
Muon Neutrino	2	1	1/2	16/5	3	Double ξ
Muon	2	1	1/2	16/5	1	Lepton number
Tau Neutrino	3	2	1/2	25/9	25/9	Double ξ
Tau	3	2	1/2	25/9	2/3	Lepton number
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	$\frac{25}{2}$	3/2	Color + Isospin
Charm	2	1	1/2	2*	2/3	Color
Strange	2	1	1/2	$\frac{26}{9}$	1	Color
Top	3	2	1/2	$\frac{1}{28}$	-1/3	Color
Bottom	3	2	1/2	$\frac{3}{2}$	1/2	Color

.7 Numerical Reconstruction

The following analysis shows the explicit calculation of all fermions with both methods:

Foundations and Experimental Input Data

Fundamental Constants:

$$\xi_0 = \xi = \frac{4}{3} \times 10^{-4} = 1.33333333 \dots \times 10^{-4} \quad (27)$$

$$v = 246 \text{ GeV} \quad (28)$$

Experimental Masses (PDG-near values):

$$m_e^{\text{exp}} = 0.0005109989461 \text{ GeV} \quad (29)$$

$$m_\mu^{\text{exp}} = 0.1056583745 \text{ GeV} \quad (30)$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \quad (31)$$

^{0*} Corrected from originally 8/9 based on detailed numerical analysis

Charged Leptons: Detailed Calculations

Electron Mass Calculation:

Direct Method:

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2), \quad f_e = 1.485 \times 10^7 \text{ GeV}^{-1} \quad (32)$$

$$= 1980 \text{ GeV}^{-1} \quad (33)$$

$$E_e = \frac{1}{\xi_e} = 0.505 \text{ MeV} \quad (34)$$

Extended Yukawa Method:

$$r_e = \frac{4}{3} \quad (35)$$

$$y_e = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} = 2.053 \times 10^{-6} \quad (36)$$

$$E_e = y_e \times 246 \text{ GeV} = 0.505 \text{ MeV} \quad (37)$$

Muon Mass Calculation:

Direct Method:

$$\xi_\mu = \frac{4}{3} \times 10^{-4} \times f_\mu(2, 1, 1/2), \quad f_\mu = 7.146 \times 10^4 \text{ GeV}^{-1} \quad (38)$$

$$= 9.528 \text{ GeV}^{-1} \quad (39)$$

$$E_\mu = \frac{1}{\xi_\mu} = 104.96 \text{ MeV} \quad (40)$$

Extended Yukawa Method:

$$y_\mu = \frac{16}{5} \times \left(\frac{4}{3} \times 10^{-4}\right)^1 = 4.267 \times 10^{-4} \quad (41)$$

$$E_\mu = y_\mu \times 246 \text{ GeV} = 104.96 \text{ MeV} \quad (42)$$

Experiment: 105.66 MeV → Deviation $\approx 0.65\%$

Experiment electron: 0.511 MeV → Deviation -1.18%

The factors f_i of the direct method follow from the fixed quantum numbers via Eq. 52: $f_e = 1/(r_e \xi^{5/2} \nu) = 1.485 \times 10^7 \text{ GeV}^{-1}$ and $f_\mu = 1/(r_\mu \xi^2 \nu) = 7.146 \times 10^4 \text{ GeV}^{-1}$. They must not be equated with the Yukawa prefactors r_i . With these the direct method gives $1/(\xi f_e) = 0.505 \text{ MeV}$ and $1/(\xi f_\mu) = 104.96 \text{ MeV}$, the same values as the Yukawa method.

Neutrino Treatment

[Neutrino Approach] This section proposes a geometric treatment of neutrino masses via a **double ξ -suppression**. According to R109, this neutrino part is not tenable (see the addendum at the end); for neutrinos, Doc. 340 is authoritative.

Neutrino Quantum Numbers

Neutrinos follow the same quantum number structure as other fermions, but with a modification due to their weak interaction nature:

Neutrino	n	l	j	Suppression
ν_e	1	0	1/2	Double ξ
ν_μ	2	1	1/2	Double ξ
ν_τ	3	2	1/2	Double ξ

Table 2: Neutrino quantum numbers with characteristic double ξ -suppression

Double ξ -Suppression Mechanism

The approach assumes that neutrinos experience an additional geometric suppression factor:

$$f(n_{\nu_i}, l_{\nu_i}, j_{\nu_i}) = f(n_i, l_i, j_i)_{\text{Lepton}} \times \xi \quad (43)$$

Neutrino Mass Calculations:

Electron Neutrino:

$$\xi_{\nu_e} = \frac{4}{3} \times 10^{-4} \times 1 \times \frac{4}{3} \times 10^{-4} = \frac{16}{9} \times 10^{-8} \quad (44)$$

$$E_{\nu_e} = \frac{1}{\xi_{\nu_e}} = 9.1 \text{ meV} \quad (45)$$

Muon Neutrino:

$$\xi_{\nu_\mu} = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} \times \frac{4}{3} \times 10^{-4} = \frac{256}{45} \times 10^{-8} \quad (46)$$

$$E_{\nu_\mu} = \frac{1}{\xi_{\nu_\mu}} = 1.9 \text{ meV} \quad (47)$$

Tau Neutrino:

$$\xi_{\nu_\tau} = \frac{4}{3} \times 10^{-4} \times \frac{25}{9} \times \frac{4}{3} \times 10^{-4} = \frac{400}{81} \times 10^{-8} \quad (48)$$

$$E_{\nu_\tau} = \frac{1}{\xi_{\nu_\tau}} = 18.0 \text{ meV} \quad (49)$$

.8 Quark Analysis with Both Methods

Explicit Calculations of Quark Masses

We use $\xi = \frac{4}{3} \times 10^{-4}$ and $v = 246 \text{ GeV}$ (SM definition from G_F , Doc. 351/R112). For the Yukawa representation:

$$y_i = r_i \xi^{p_i}, \quad m_i^{\text{pred}} = y_i v.$$

For the direct geometric representation:

$$f_i = \frac{1}{\xi m_i^{\text{exp}}}, \quad m_i^{\text{exp}} = \frac{1}{\xi f_i}.$$

Quark	p_i	r_i (corr.)	m_i^{pred} (GeV)	m_i^{exp} (GeV)	rel. error (%)	Remark
Up	3/2	6	2.272×10^{-3}	2.16×10^{-3}	+5.21	OK
Down	3/2	25/2	4.734×10^{-3}	4.70×10^{-3}	+0.73	OK
Strange	1	26/9	9.476×10^{-2}	9.35×10^{-2}	+1.34	OK
Charm	2/3	2	1.284×10^0	1.273	+0.87	Corrected
Bottom	1/2	3/2	4.261×10^0	4.18	+1.93	OK
Top	-1/3	1/28	1.7198×10^2	172.57	-0.35	OK

Table 3: Yukawa predictions with corrected r_i, p_i and comparison with the PDG 2024 masses.

Correction for the Charm Quark

The originally specified value $r_c = 8/9$ does not reproduce the reference mass $m_c = 1.273$ GeV (PDG 2024). The required value is:

$$r_c^{\text{required}} = \frac{m_c^{\text{exp}}}{v\xi^{2/3}} \approx 1.983 \approx 2.$$

Therefore, $r_c \approx 2$ was used in the corrected universal table. With $r_c = 2$ exactly, $m_c = 1.284$ GeV (+0.87 %).

.9 Experimental Comparison

Accuracy Analysis

Comparison of the T0 values with the experimental masses:

Particle	T0 Prediction	Experiment	Accuracy	Type
<i>Charged Leptons</i>				
Electron	0.505 MeV	0.511 MeV	98.82%	Lepton
Muon	104.96 MeV	105.66 MeV	99.35%	Lepton
Tau	1783.4 MeV	1776.86 MeV	99.63%	Lepton
<i>Neutrinos</i>				
ν_e	9.1 meV	< 450 meV	Compatible	Neutrino
ν_μ	1.9 meV	< 180 keV	Compatible	Neutrino
ν_τ	18.0 meV	< 18 MeV	Compatible	Neutrino
<i>Quarks</i>				
Up quark	2.272 MeV	2.16 MeV	94.79%	Quark
Down quark	4.734 MeV	4.70 MeV	99.27%	Quark
Strange quark	94.76 MeV	93.5 MeV	98.66%	Quark
Charm quark	1.284 GeV	1.273 GeV	99.13%	Quark
Bottom quark	4.261 GeV	4.18 GeV	98.07%	Quark
Top quark	171.99 GeV	172.57 GeV	99.65%	Quark
Average			98.6%	All Fermions

Table 4: Comparison of T0 values with experimental masses**Result for Charged Leptons and Quarks**

The T0-Model achieves 98.6% average accuracy across the charged leptons and quarks, with the single parameter ξ and fixed rational quantum numbers. The neutrino rows are given only as compatibility with bounds; the neutrino part is not tenable according to R109 (see addendum).

.10 Predictive Power of the Established System**New Particle Generations**

Extrapolating the established patterns to a 4th generation gives:

4th Generation (extrapolated):

$$n = 4, \quad \pi_4 = \frac{1}{2}, \quad r_4 \approx 2.0 \quad (50)$$

$$m_{4\text{thGen}} = r_4 \times \xi^{1/2} \times v \approx 5.7 \text{ GeV} \quad (51)$$

The calculation gives 5.68 GeV. A charged lepton of a 4th generation is excluded below about 100 GeV by LEP, and the Z width allows only three light neutrinos; this extrapolation is therefore not a viable prediction.

Quark Sector Extrapolation

The lepton patterns can be transferred to quarks:

Quark	Generation	r_i	π_i	Prediction
Up	1	6	3/2	2.3 MeV
Down	1	12.5	3/2	4.7 MeV
Charm	2	2.0	2/3	1.3 GeV
Strange	2	2.89	1	95 MeV
Top	3	0.036	-1/3	173 GeV
Bottom	3	1.5	1/2	4.3 GeV

Table 5: Quark predictions from established patterns

.11 Corrected Interpretation of Mathematical Equivalence

Meaning of the Equivalence

The mathematical equivalence of both methods is **given by definition** when the parameters (r_i or f_i) are determined from the same experimental masses. The equivalence is not proof of the theory, but a consistency property of the mathematical structure.

Transformation Relationship as Bridge

The relationship:

$$f_i = \frac{1}{r_i \xi_{\pi_i} v \xi_0} \quad (52)$$

connects both methods mathematically. If r_i is determined from experimental masses, f_i follows automatically and vice versa.

Particle	m^{exp} (GeV)	r_i (Yukawa)	f_i (direct)	Accuracy
Electron	0.000511	4/3	1.485×10^7	98.82%
Muon	0.10566	16/5	7.146×10^4	99.34%
Tau	1.77686	25/9	4.205×10^3	99.63%
ν_e	9.1×10^{-6}	1.349	8.235×10^{10}	Prediction
ν_μ	1.9×10^{-6}	3.221	3.947×10^{11}	Prediction
ν_τ	18.0×10^{-6}	2.778	3.989×10^{10}	Prediction

Table 6: Numerical equivalence of both T0 methods for all leptons

For the charged leptons the fixed quantum numbers 4/3, 16/5, 25/9 are given with the f_i following from them via Eq. 52 (in GeV^{-1}); both methods give the same masses and hence the same accuracy. The neutrino rows belong to the neutrino part, which is not tenable (see addendum, R109).

.12 Experimental Predictions and Precision Tests

Modified QED Vertex Corrections

T0 theory predicts modified Feynman rules:

$$\text{Time field vertex: } -i\gamma^\mu T_\mu^{(T)} = i\gamma^\mu \frac{\partial_\mu m}{m^2} \quad (53)$$

$$\text{Modified fermion propagator: } S_F^{(T0)}(p) = S_F(p) \cdot \left[1 + \frac{\beta}{p^2} \right] \quad (54)$$

Neutrino Validation

The T0 neutrino values are compatible with the experimental bounds listed here (on the tenability of the neutrino part, see addendum, R109):

Parameter	T0 Prediction	Experimental Limit	Status
m_{ν_e}	9.1 meV	< 450 meV (KATRIN)	Satisfied
m_{ν_μ}	1.9 meV	< 180 keV (indirect)	Satisfied
m_{ν_τ}	18.0 meV	< 18 MeV (indirect)	Satisfied
$\sum m_\nu$	29.8 meV	< 60 meV (Cosmology 2024)	Satisfied

Table 7: T0 neutrino predictions vs. experimental constraints

Neutrino Mass Hierarchy

The T0-Model predicts **normal ordering**: $m_{\nu_\mu} < m_{\nu_e} < m_{\nu_\tau}$, which is consistent with current oscillation data preferences.

.13 Methodological Foundation

Reversibility of the Established System

After the establishment phase, the T0 system can make predictions:

Established Lepton Patterns:

$$\text{1st Generation (n=1): } \pi_i = \frac{3}{2}, \quad r_e = \frac{4}{3} \approx 1.333 \quad (55)$$

$$\text{2nd Generation (n=2): } \pi_i = 1, \quad r_\mu = \frac{16}{5} = 3.2 \quad (56)$$

$$\text{3rd Generation (n=3): } \pi_i = \frac{2}{3}, \quad r_\tau = \frac{25}{9} \approx 2.778 \quad (57)$$

Experimental Testability

The T0 predictions are experimentally falsifiable:

1. **Precision Measurements:** Refinement of r_i parameters
2. **Neutrino Tests:** Direct neutrino mass measurements
Methodologically, the following speaks for the T0 procedure:
 1. **Systematic Structure:** All parameters follow recognizable patterns

2. **Predictive Power:** After establishment, new particles become predictable
3. **Experimental Testability:** Predictions are falsifiable
4. **QFT Foundation:** Quantum field theoretical derivation of the ξ -constant
5. **Historical Models:** The methodology of pattern physics has been successful several times

.14 Parameter Economy and Universal Structure

No Adjustable Parameters

The scale of the T0 coefficients is determined by ξ , which can also be checked against Higgs parameters (deviation from $4/3 \times 10^{-4}$ about -2.4%):

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \quad (58)$$

The relation is a consistency check, not a derivation (see Section .5). Together with the rational quantum numbers (r_i, p_i) , no continuously adjustable parameter remains; v serves as a conversion anchor.

Universal Quantum Number Table

Particle	n	l	j	r_i	p_i	Special
<i>Charged Leptons</i>						
Electron	1	0	1/2	4/3	3/2	–
Muon	2	1	1/2	16/5	1	–
Tau	3	2	1/2	25/9	2/3	–
<i>Neutrinos</i>						
ν_e	1	0	1/2	4/3	5/2	Double ξ
ν_μ	2	1	1/2	16/5	3	Double ξ
ν_τ	3	2	1/2	25/9	25/9	Double ξ
<i>Quarks</i>						
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	25/2	3/2	Color + Isospin
Charm	2	1	1/2	2	2/3	Color
Strange	2	1	1/2	26/9	1	Color
Top	3	2	1/2	1/28	-1/3	Color
Bottom	3	2	1/2	3/2	1/2	Color

Table 8: Universal quantum number table for all fermions

Winding numbers and topological assignment

The rational quantum numbers (r_i, p_i) are not arbitrary. They follow from the winding numbers (n_θ, n_ϕ) of the 4D torus T^4 via $r_i = n_\phi^2 / n_\theta$ (Doc. 317, August 2026):

Lepton	Gen.	n_θ	n_ϕ	r_i	p_i	KSAU knot
e	1	3	2	$4/3$	$3/2$	3_1 (trefoil)
μ	2	5	4	$16/5$	1	6_3 (amphicheiral)
τ	3	9	5	$25/9$	$2/3$	7_1 (torus knot)
ν_e	1	3	2	$4/3$	$5/2$	—
ν_μ	2	5	4	$16/5$	3	—
ν_τ	3	9	5	$25/9$	$25/9$	—

The exponent step $\Delta p = 1/3$ corresponds to the modulation frequency of the torus winding. The knot assignment (KSAU, Zenodo 2026) is **[S]**; the FFGFT quantum numbers are **[K]**. Details in Doc. 317 (KSAU-FFGFT synthesis).

.15 Critical Evaluation and Limitations

Theoretical Open Questions

1. **Number of Generations:** Why exactly three generations?
2. **Hierarchy Problem:** Connection between different energy scales
3. **CP Violation:** Incorporation of CKM and PMNS mixing matrices

.16 Concluding Assessment

Scientific Status

The T0-Model offers a systematic description of particle masses. The combination of:

- **High numerical accuracy** (98.6% across charged leptons and quarks)
- **Parameter economy** (one parameter ξ , rational quantum numbers)
- **Broad coverage** (charged leptons and quarks; neutrinos see addendum)
- **QFT consistency** (1-loop derivation of the ξ -constant)
- **Experimental testability** (specific falsifiable predictions)
makes the model a testable proposal.

Significance for Fundamental Physics

If experimentally confirmed, the T0-Model would considerably change our understanding of particle physics:

1. **Geometric Interpretation:** Particle masses as manifestations of 3D spatial geometry
2. **Unification:** All fermions follow the same universal structure
3. **Predictive Power:** New particles become predictable from established patterns
4. **Simplicity:** Strong simplification of complex phenomena

The T0-Model is an example showing that a unified description does not necessarily require greater complexity; the basic structure might be simple.

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Addendum: Status Clarification (7 September 2026)

R105: Retroactive marker certification (1 September 2026). This document was created before the marker system was introduced (Doc. A010) and carries no inline markers. Its derivations are internally consistent and serve as chain links for the Galois programme (Docs. 336, 338, 339).

Retroactive status for this document: Winding numbers r_i , exponents p_i , and the Yukawa method $m_i = r_i \xi^{p_i} v$ for charged leptons and quarks: **[K]**.

Dependent documents may treat these results as a secured chain link via reference to R105, without modifying this document.

R109: Restriction — neutrino part not viable [X] (1 September 2026). The “direct method” $E_i = 1/\xi_i$ for neutrinos is not a calculation: $1/(\xi r_i)$ gives the ratio 1 : 0.42 for $e : \mu$ (inverse hierarchy, dimensionless); the MeV values alongside are measurements. The neutrino masses 9.1/1.9/18.0 meV follow from three mutually incompatible prescriptions; the exponent $p_{\nu_\tau} = 25/9$ is r_{τ} , erroneously copied into the exponent column.

Valid: the Yukawa method $m_i = r_i \xi^{p_i} \nu$ for charged leptons and quarks ($< 1.2\%$ deviation). The **[K]** certification from R105 applies to this document *only* for r_i , p_i , and the Yukawa method — not for the neutrino part and not for the claimed equivalence of both methods.

Clarification: Both methods are exactly equivalent via $f_i = 1/(r_i \xi^{p_i+1} \nu)$ (Eq. 52) and give the same masses with fixed r_i , p_i .

Authoritative for neutrinos: Doc. 340 (R103, R108).